

# DW\_lp\_piped\_mult

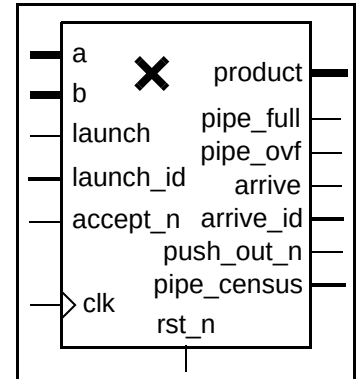
## Low Power Pipelined Multiplier

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

### Features and Benefits

- Built-in pipelining and power management
- Automatically enables register retiming
- Parameterized operand widths
- Parameterized pipeline stages
- Launch identifier tracking propagation
- Operand isolation capability on a and b

### Revision History



### Description

DW\_lp\_piped\_mult performs a pipelined multiply based on two operands with the added benefit of power savings. Pipeline control is integrated and applied to pipelined register levels (when configured in) to minimize power consumption. Pipeline register re-timing is automatically enabled for balancing between logic stages.

**Table 1-1 Pin Descriptions**

| Pin Name  | Width                                | Direction | Function                                                                                                      |
|-----------|--------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------|
| clk       | 1 bit                                | Input     | Input clock                                                                                                   |
| rst_n     | 1 bit                                | Input     | Reset, active-low (not used if parameter rst_mode=0)                                                          |
| a         | <i>a_width</i> bits                  | Input     | Multiplier                                                                                                    |
| b         | <i>b_width</i> bits                  | Input     | Multiplicand                                                                                                  |
| product   | <i>a_width</i> + <i>b_width</i> bits | Output    | Product $a \times b$                                                                                          |
| launch    | 1 bit                                | Input     | Control to begin a new multiply operation                                                                     |
| launch_id | <i>id_width</i> bits                 | Input     | Identifier for the corresponding asserted <code>launch</code>                                                 |
| pipe_full | 1 bit                                | Output    | Upstream notification that pipeline is full                                                                   |
| pipe_ovf  | 1 bit                                | Output    | Status Flag indicating pipe overflow                                                                          |
| accept_n  | 1 bit                                | Input     | <code>product</code> result accepted from downstream logic (active low)                                       |
| arrive    | 1 bit                                | Output    | <code>product</code> result is valid                                                                          |
| arrive_id | <i>id_width</i> bits                 | Output    | <code>launch_id</code> from the originating <code>launch</code> that produced the <code>product</code> result |

**Table 1-1 Pin Descriptions (Continued)**

| Pin Name    | Width  | Direction | Function                                                                                                                                                                                                                                                                                |
|-------------|--------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| push_out_n  | 1 bit  | Output    | Push performed to downstream FIFO element (active low)                                                                                                                                                                                                                                  |
| pipe_census | M bits | Output    | Number of pipeline register levels currently occupied<br>Note: The value of M is equal to the larger of '1' or $\text{ceil}(\log_2(\text{in\_reg} + \text{stages} + \text{out\_reg}))$<br>Example: if $\text{in\_reg} = 1$ , $\text{stages} = 2$ , $\text{out\_reg} = 1$ , then $M = 2$ |

**Table 1-2 Parameter Description**

| Parameter | Values                  | Description                                                                                                                         |
|-----------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| a_width   | $\geq 1$<br>Default: 8  | Word length of a                                                                                                                    |
| b_width   | $\geq 1$<br>Default: 8  | Word length of b                                                                                                                    |
| id_width  | 1 to 1024<br>Default: 8 | Width of <code>launch_id</code>                                                                                                     |
| in_reg    | 0 or 1<br>Default: 0    | Input register control <ul style="list-style-type: none"> <li>0: No input register</li> <li>1: Include input register</li> </ul>    |
| stages    | 1 to 1022<br>Default: 4 | Number of pipeline stages                                                                                                           |
| out_reg   | 0 or 1<br>Default: 0    | Output register control <ul style="list-style-type: none"> <li>0: No output register</li> <li>1: Include output register</li> </ul> |
| tc_mode   | 0 or 1<br>Default: 0    | Two's complement control <ul style="list-style-type: none"> <li>0: Unsigned</li> <li>1: Signed</li> </ul>                           |
| rst_mode  | 0 to 1<br>Default: 0    | Reset mode <ul style="list-style-type: none"> <li>0: Asynchronous reset</li> <li>1: Synchronous reset</li> </ul>                    |

**Table 1-2 Parameter Description (Continued)**

| Parameter   | Values               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| op_iso_mode | 0 to 4<br>Default: 0 | <p>Operand isolation mode (controls datapath gating for minPower flow)<br/>Allows you to set the style of minPower datapath gating for this module</p> <ul style="list-style-type: none"> <li>0: Use the DW_lp_op_iso_mode<sup>a</sup> synthesis variable</li> <li>1: 'none'</li> <li>2: 'and'</li> <li>3: 'or'</li> <li>4: Preferred gating style: 'and'</li> </ul> <p>Datapath gating is inserted only when there are no input registers on the operands at the component boundary. When inserted, datapath gating circuits are placed immediately after the input ports of the component (see <a href="#">Figure 1-2</a> on page 5).</p> |

a. The DW\_lp\_op\_iso\_mode synthesis variable is available only in Design Compiler.

DW\_lp\_op\_iso\_mode sets a global style of datapath gating. To use the global style, set *op\_iso\_mode* to '0'. Note that if the *op\_iso\_mode* parameter is set to '0' and DW\_lp\_op\_iso\_mode is either not set or set to 0, then no datapath gating is inserted for this component.

**Table 1-3 Synthesis Implementations**

| Implementation Name | Function        | License Feature Required                                                                                                                 |
|---------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| rtl                 | Synthesis model | <ul style="list-style-type: none"> <li>DesignWare (P-2019.03 and later)</li> <li>DesignWare-LP<sup>a</sup> (before P-2019.03)</li> </ul> |

a. For Design Compiler versions before P-2019.03, see [“Enabling minPower”](#) on page 13.

**Table 1-4 Simulation Models**

| Model                                | Function                             |
|--------------------------------------|--------------------------------------|
| DW03.DW_LP_PIPED_MULT_CFG_SIM        | Design unit name for VHDL simulation |
| dw/dw03/src/DW_lp_piped_mult_sim.vhd | VHDL simulation model source code    |
| dw/sim_ver/DW_lp_piped_mult.v        | Verilog simulation model source code |

## Functional Description

### Block Diagram

Figure 1-1 shows the block diagram of the DW\_lp\_piped\_mult component:

Figure 1-1 DW\_lp\_piped\_mult Basic Block Diagram

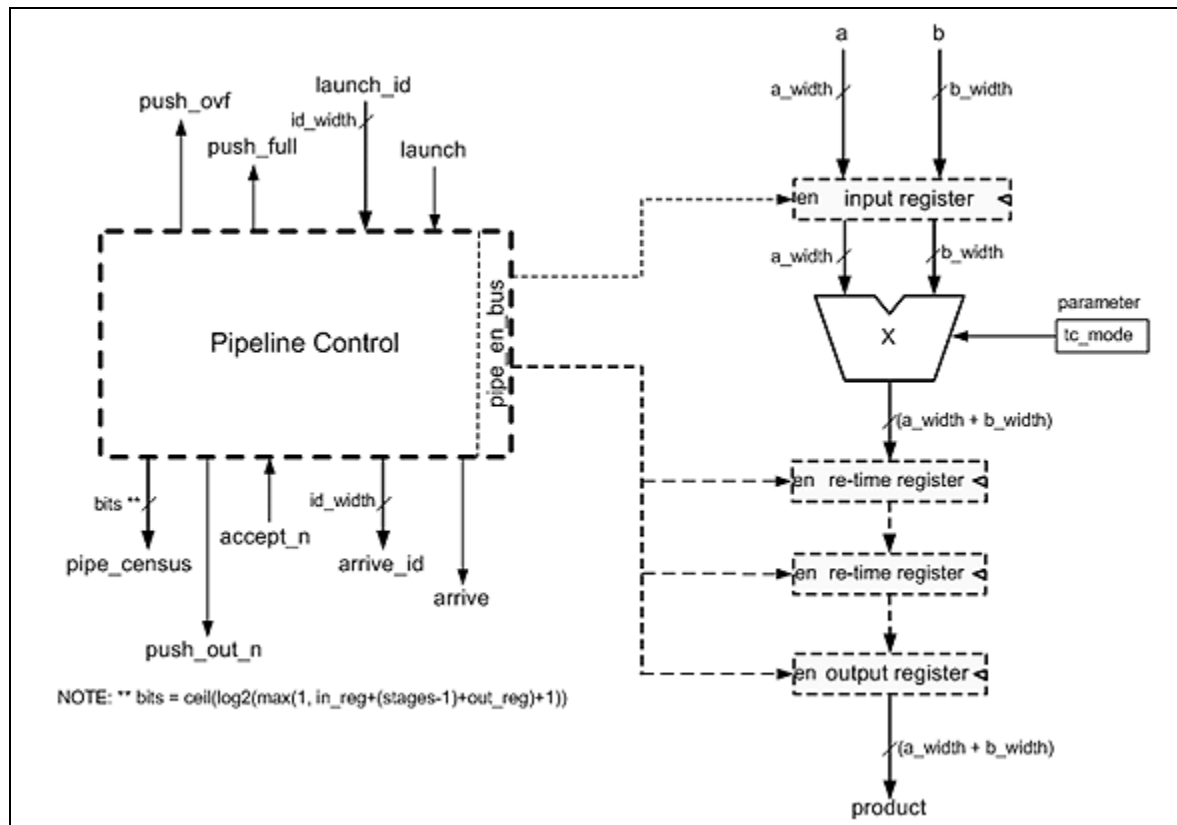
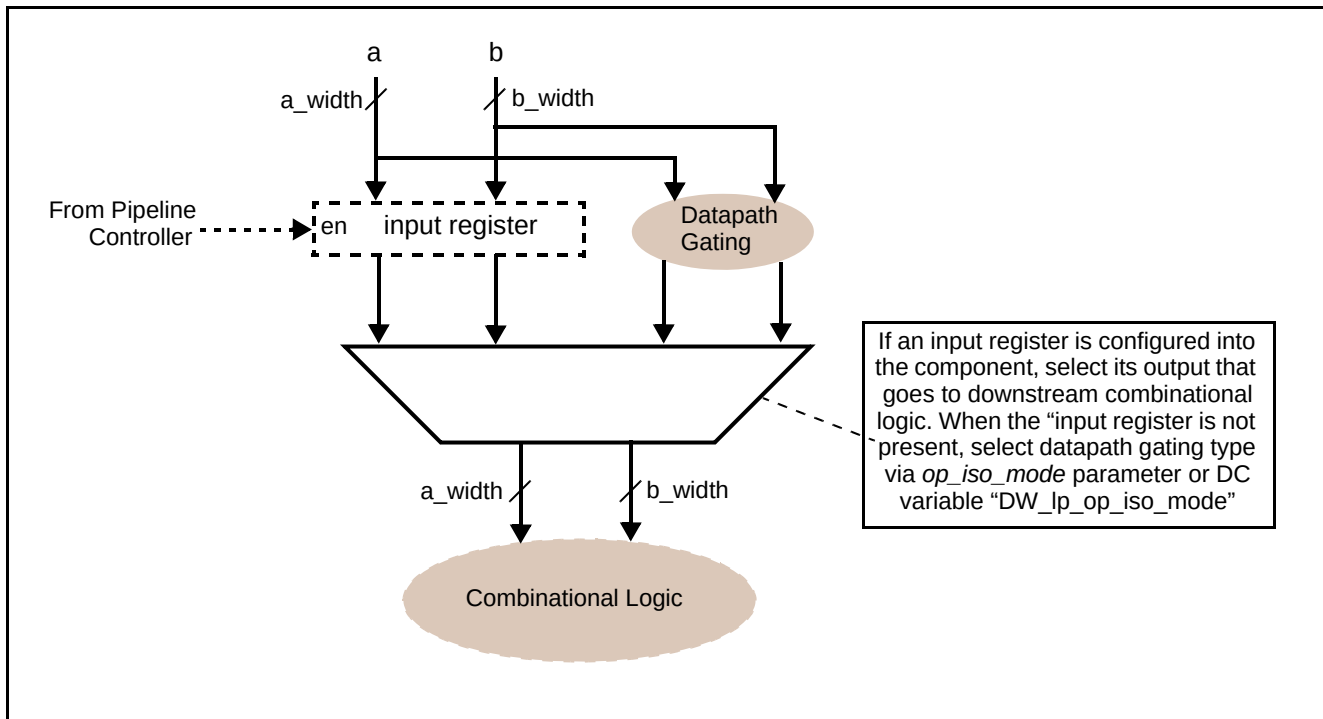


Figure 1-2 shows where datapath gating is inserted when the *op\_iso\_mode* parameter enables it.

**Figure 1-2 Location of Datapath Gating (If Inserted)**



## Pipelining

The DW\_lp\_piped\_mult is configurable embed pipeline register levels. Setting the value for the parameters *in\_reg*, *stages*, and *out\_reg* (see Table 1-5 on page 6) determines the number of pipeline register level(s) that are inserted. Therefore, depending on the parameter *in\_reg*, *stages*, and *out\_reg* settings, the number of clock cycles for the product result to propagate varies.

This DW\_lp\_piped\_mult is designed to make it easy to pipeline the multiplier using the register retiming features of Design Compiler (DC). It also contains parameter controlled input and output registers which will stay in place at their respective block boundary, that is, they are not allowed to be moved by DC register retiming features. The input and output registers are not available when using DC versions earlier than A-2007.12.

The parameter *stages* refers to the number of logic stages desired after register retiming is performed. The number of register levels is not necessarily the same as the number of logic stages. If no input or output registers are used (i.e. *in\_reg* = 0 or *out\_reg* = 0), then there is one fewer register level than logic *stages*. If either an input register or output register is specified, then the number of register levels is the same as the number of logic *stages*. If both input and output registers are specified, then the number of register levels is the number of logic *stages* + 1. This is described in Table 1-5. The number of pipeline register levels that can be retimed is always *stages* - 1.

**Table 1-5 Number of Pipeline Register Levels**

| in_reg | out_reg | Number of Pipeline Register Levels |
|--------|---------|------------------------------------|
| 0      | 0       | stages - 1                         |
| 0      | 1       | stages                             |
| 1      | 0       | stages                             |
| 1      | 1       | stages + 1                         |

## Pipeline Control and Power Savings

Running in parallel to the pipeline register levels is pipeline control logic (as seen in [Figure 1-1](#)) that monitors the activity. In cases where there is inactivity on a particular register level of the pipeline, the pipeline control disables those levels to promote power savings. Furthermore, if using the Synopsys Power Compiler tool, the presence of the pipeline control and its wiring to the pipeline register levels provides an opportunity for increased power reduction in the form of clock gating.

Along with the potential power savings that the pipeline control provides, it can be utilized to improve performance in cases where intermittent launch operations are present and there contains first-in first-out (FIFO) structures upstream and downstream of the DW\_lp\_piped\_mult. The handshake is made between the DW\_lp\_piped\_mult and the external FIFOs via the `accept_n` and `pipe_full` ports. Effectively, the DW\_lp\_piped\_mult can be considered part of the external FIFO structures. The performance gain comes when inactive (bubbles) stages are detected. These pipeline 'bubbles' are removed to produce a contiguous set of active pipeline stages. The result is empty pipeline slots at the head of (or entering) the DW\_lp\_piped\_mult pipeline for new operations to be launched. Advancing the shifting of operations through the pipeline when a valid product result is available (`arrive = 1`) is controlled by the `accept_n` input. When the multiplier pipeline is full of active entries, the `pipe_full` output is 1. To disable this feature in cases where no external FIFOs are present, set the `accept_n` input to 0 which will effectively eliminate any flow control. At the same time, the `pipe_full` output would always be 0.

To assist in tracking of 'launched' operands, the pipeline control logic provides interface ports called `launch_id` and `arrive_id`. The `launch_id` input is assigned a value during an active launch operation. Given that `launch_id` values are unique in successive launch operations, the product results can be distinguished from one another with the assertion of `arrive` and the associated `arrive_id`. The `arrive_id` is the `launch_id` from the originating launch that produced the valid product result.

## No Pipeline Register Levels Specified

In cases where no pipelining is required through the DW\_lp\_piped\_mult (`in_reg = 0`, `stages = 1`, and `out_reg = 0`), the pipeline control flow control handshaking/status signals still remain active and meaningful with one exception. The `pipe_census`, which is intended to count the number of active pipeline register levels, becomes irrelevant and is fixed to 0. For timing waveforms, see [Figure 1-6](#) on page [11](#).

## Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW\_SUPPRESS\_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)
```

The warning messages for this model include the following:

- If values other than 1 or 0 are present on a clock port, the following message is displayed:

```
WARNING: <instance_path>.<clock_name>_monitor:  
at time = <timestamp>, Detected unknown value, x, on <clock_name> input.
```

To suppress only this warning message for all DWBB components, use the following macro:

- Define the DW\_DISABLE\_CLK\_MONITOR macro. You can define this macro in the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_DISABLE_CLK_MONITOR
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_DISABLE_CLK_MONITOR (which is used for the Synopsys VCS simulator)
```

This message is also suppressed using the DW\_SUPPRESS\_WARN macro explained earlier.

## Timing Waveforms

Figure 1-3 shows a case where there are two pipeline register levels since  $in\_reg = 0$ ,  $stages = 2$ , and  $out\_reg = 1$ . Launching is performed while  $accept\_n$  is de-asserted causing the pipeline to fill up. This is indicated by  $pipe\_full$  going to '1' while  $accept\_n = 1$ . The  $pipe\_census[1:0]$  value is '2' which indicates that all the pipeline register levels contain active results. Notice that the first  $product[15:0]$  result of '1416' is available as indicated by  $arrive$  being '1'. Also, the identification tracking from the  $launch\_id[3:0]$  of '1' is seen upon arrival with the matching value on  $arrive\_id[3:0]$ .

At the point that the pipeline is full,  $accept\_n$  is asserted ('0') to begin emptying the pipeline. Note that  $pipe\_full$  de-asserts when  $accept\_n$  is asserted, but the  $pipe\_census[1:0]$  value still indicates '2' the next clock cycle since a launch coincided with the asserted  $accept\_n$ . Once the launching activity ceases, the continued assertion of  $accept\_n$  drains the pipeline of active  $product[15:0]$  results with  $pipe\_census[1:0]$  eventually going to '0'.

The DW\_lp\_piped\_mult is configured to operate in two's complement mode.

**Figure 1-3 Launching Until Full, Accepting Until Empty**

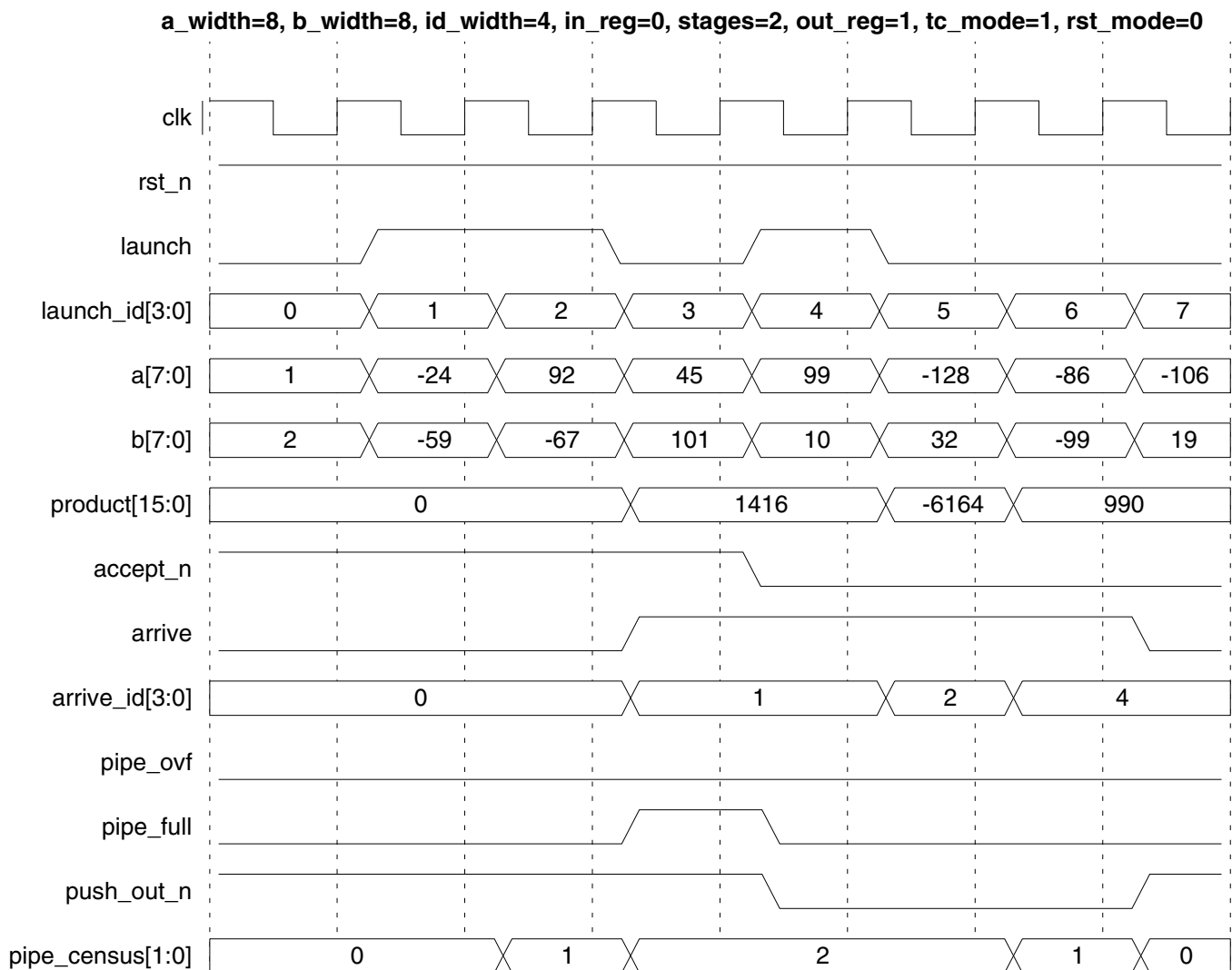


Figure 1-4 shows a case where `launch` is asserted every other clock cycle while `accept_n` is always asserted ('0'). There are 4 pipeline register levels. So, the first product result of '4' (`arrive_id[2:0]` of '1') arrives after the fourth rising-edge of `clk` from the first asserted `launch` with accompanying `launch_id[2:0]` of '1'. Any values of `a[4:0]` and `b[3:0]` are ignored when `launch` is '0'.

**Figure 1-4 Launch Every Other Cycle with Asserted `accept_n`**

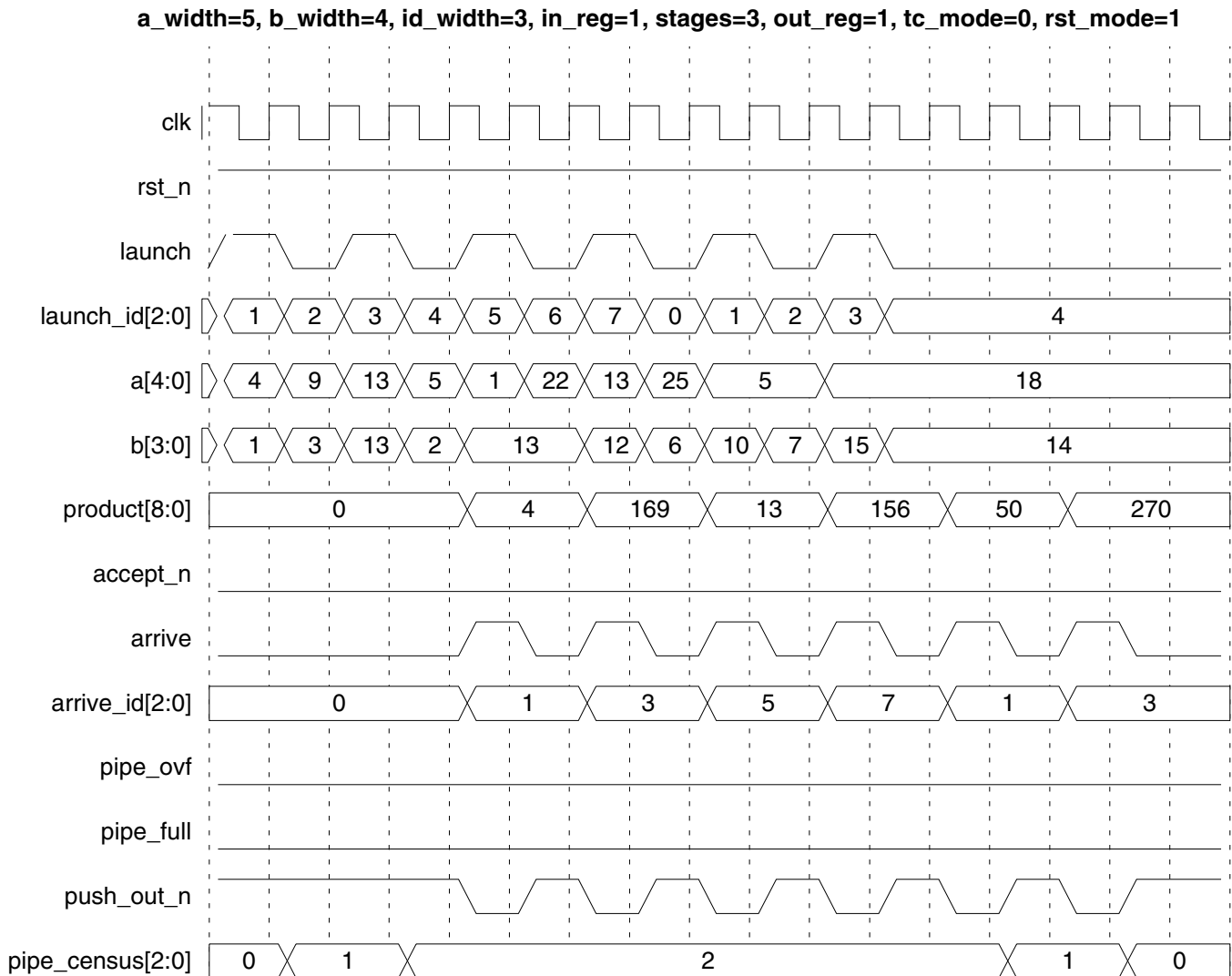


Figure 1-5 depicts a pipeline overflow condition. This is the same configuration as shown in Figure 1-4 on page 9. The `pipe_ovf` output is registered and gets asserted following the rising-edge of `clk` when the pipeline is full (`pipe_full` is '1'), `launch` is asserted ('1'), and `accept_n` is not asserted ('1'). In this situation, the launched operation is ignored and the pipeline contents are preserved.

Figure 1-5 Pipeline Overflow

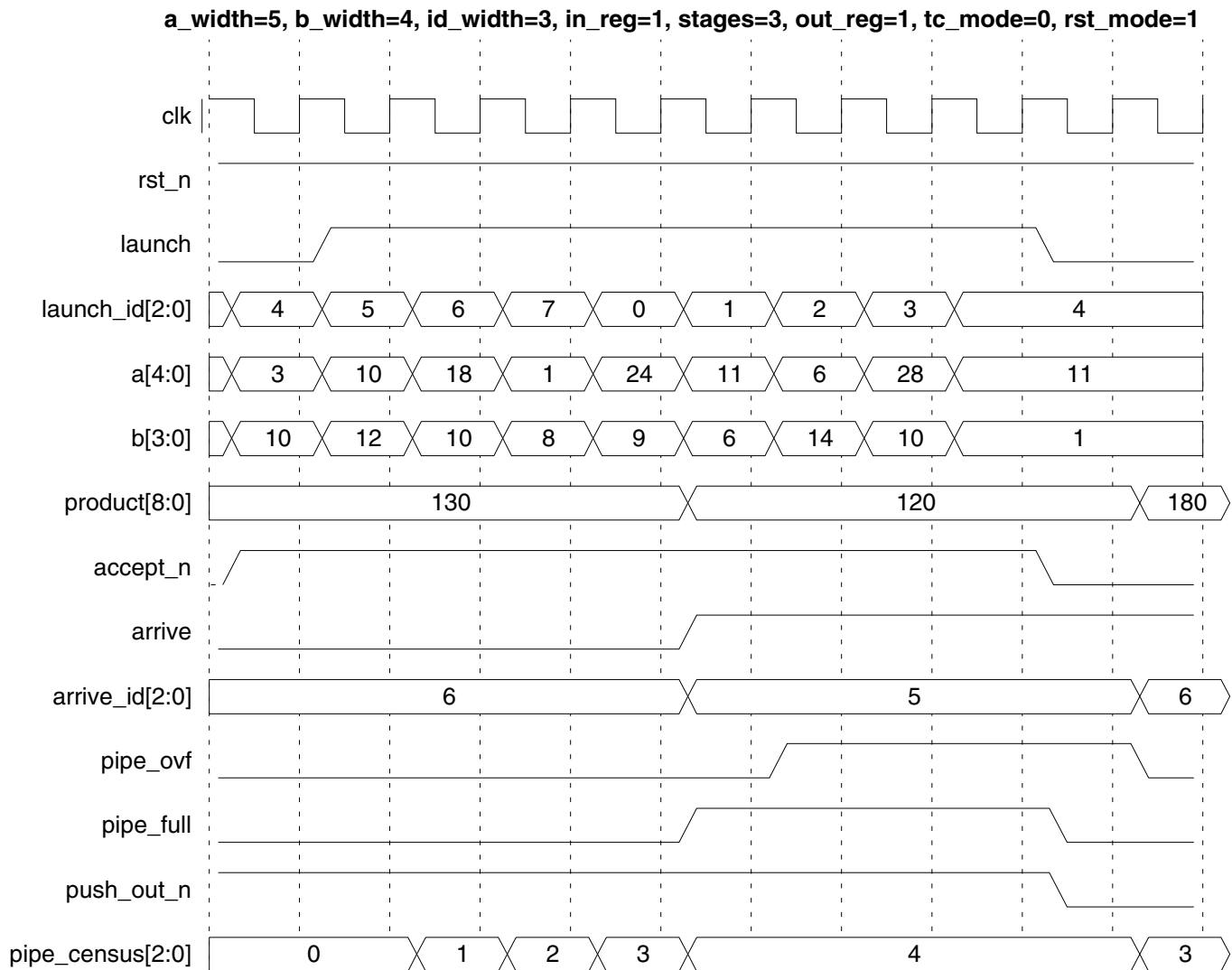


Figure 1-6 depicts a scenario when there is no pipelining configured into the DW\_lp\_piped\_mult which is defined when  $in\_reg = 0$ ,  $stages = 1$ , and  $out\_reg = 0$ . Thus, the product result is a pure combinational logic path from  $a[3:0]$  and  $b[3:0]$ . The flow control/status outputs arrive, arrive\_id[7:0], pipe\_full, pipe\_ovf, and push\_out\_n still have meaning. However, the output pipe\_census has no meaning since no pipeline register levels exist. Hence, pipe\_census will always be driven to '0'.

Notice that when launch is asserted and accept\_n is not, the register output pipe\_ovf goes to '1'. This is due to the fact that when  $accept\_n = 1$ , it implies that the downstream device cannot accept any more results. Thus, a launch under this condition will result in overrun and the subsequent product result is lost.

**Figure 1-6 No Pipeline Specified ( $in\_reg = 0$ ,  $stages = 1$ ,  $out\_reg = 0$ )**

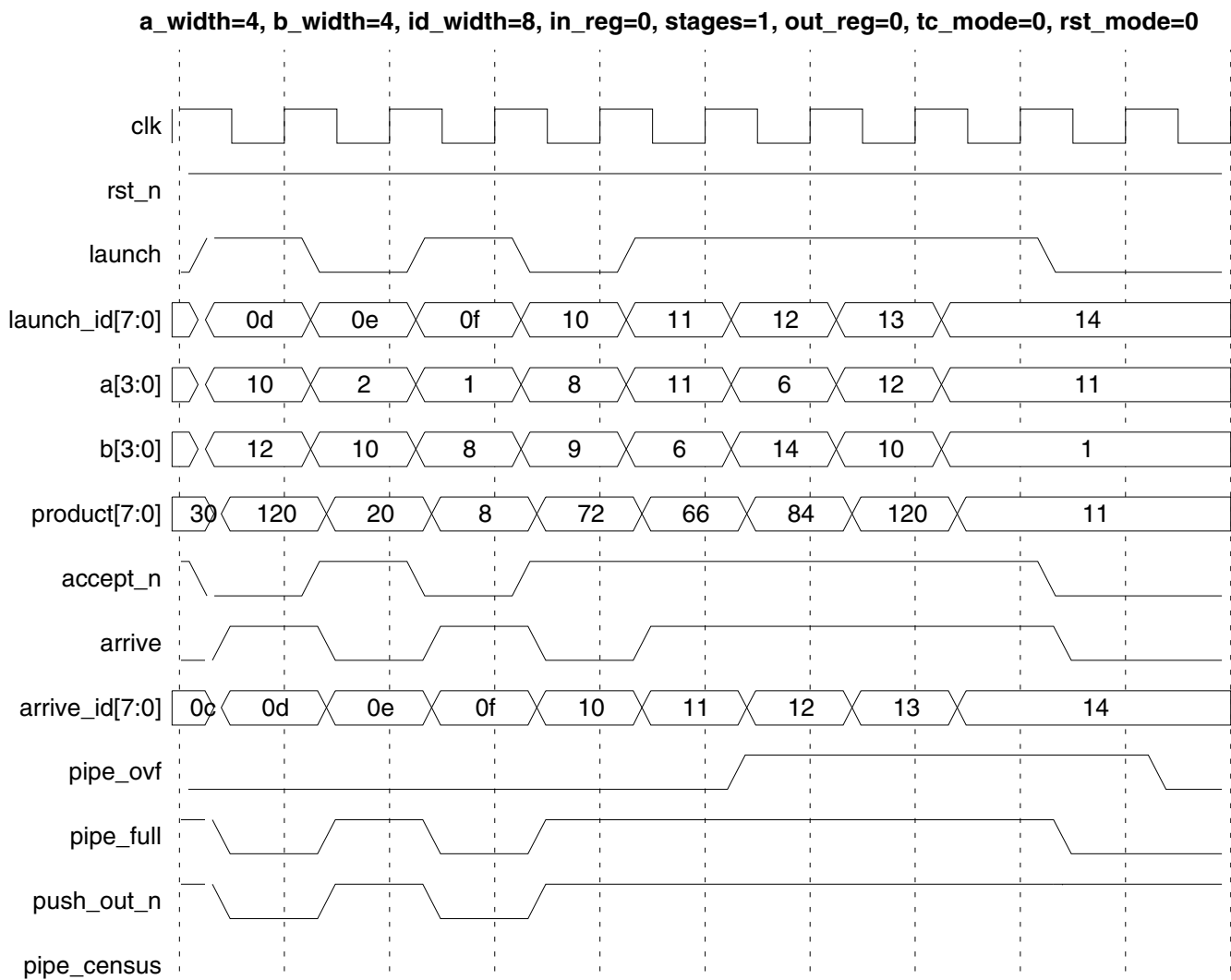


Figure 1-7 shows the affects that the assertion of `rst_n` while configured for asynchronous resetting (`rst_mode = 0`).

**Figure 1-7 No Pipeline Specified (`in_reg = 0`, `stages = 1`, `out_reg = 0`)**

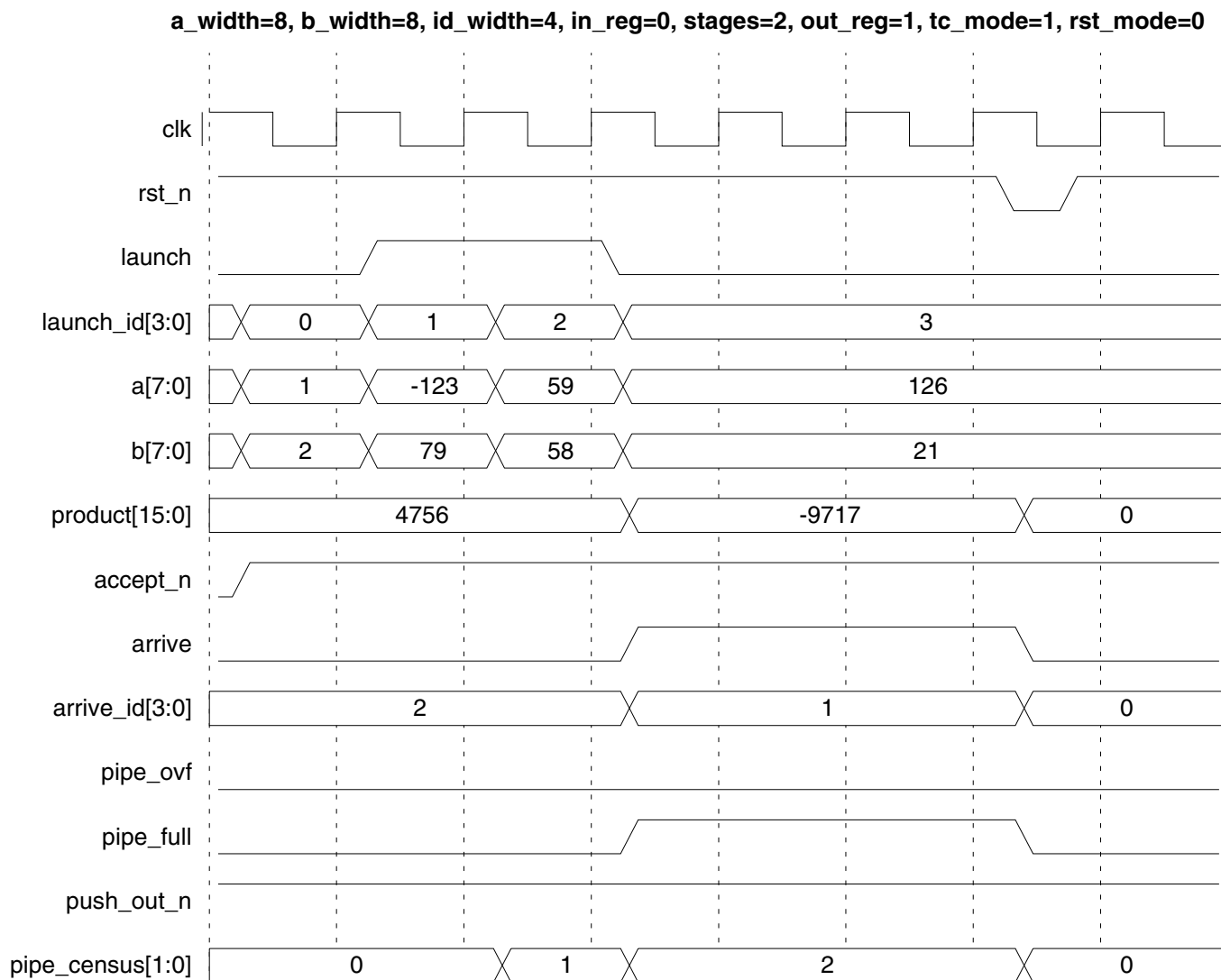
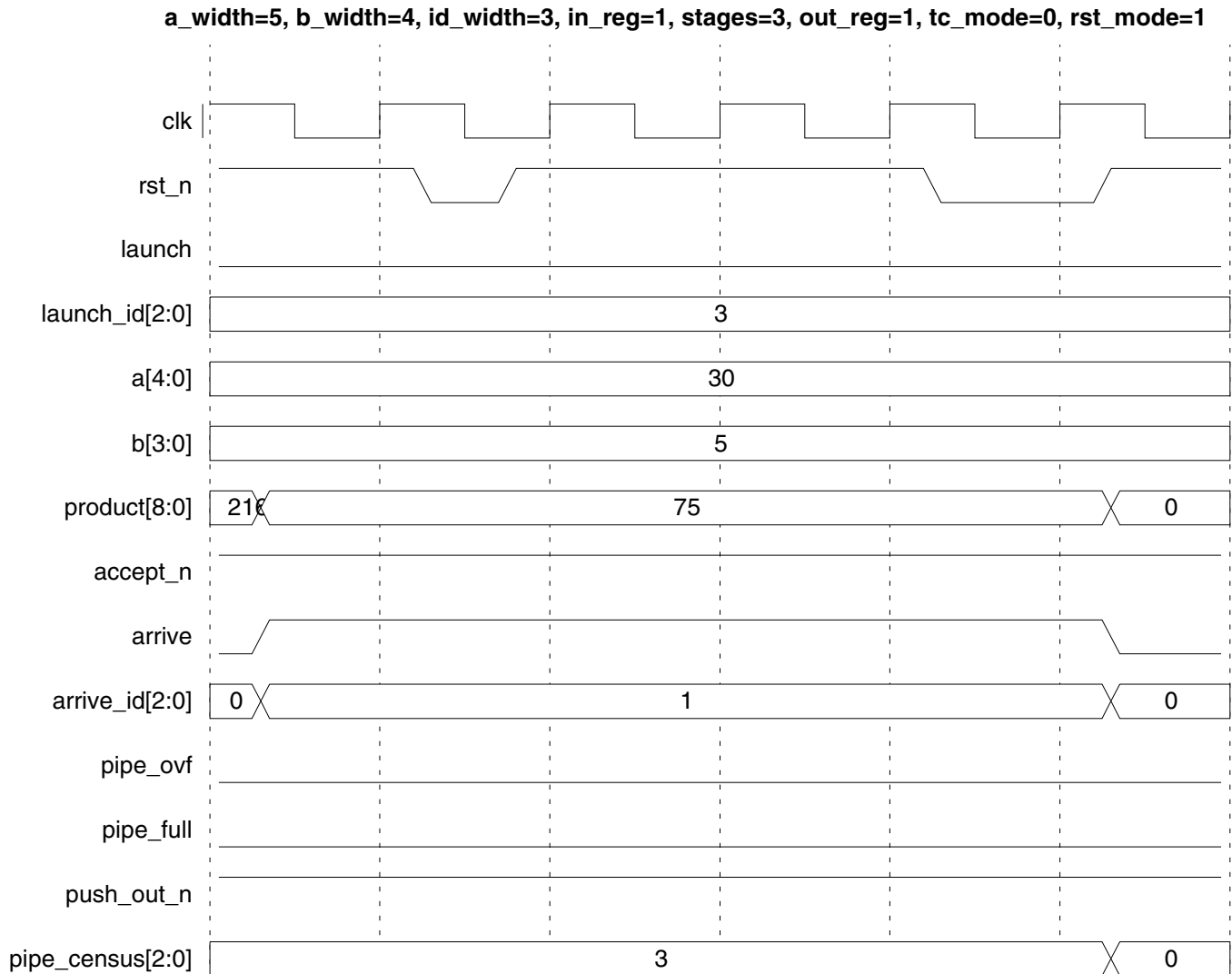


Figure 1-8 shows the affects of asserting `rst_n` while configured for synchronous resetting (`rst_mode = 1`). Only when a '0' state of `rst_n` is sampled by the rising-edge of `clk` does the clearing of register elements occur.

**Figure 1-8 Synchronous Reset Behavior (`rst_mode = 1`)**



## Enabling minPower

In Design Compiler (version P-2019.03 and later) and Fusion Compiler, you can instantiate this component and use all its features without special settings.

For versions of Design Compiler before P-2019.03, enable minPower as follows:

```
set synthetic_library {dw_foundation.sldb dw_minpower.sldb}
set link_library {* $target_library $synthetic_library}
```

---

## Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Blocks User Guide](#)

## HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp.all;

entity DW_lp_piped_mult_inst is
    generic (
        inst_a_width : POSITIVE := 8;
        inst_b_width : POSITIVE := 8;
        inst_id_width : POSITIVE := 16;
        inst_in_reg : NATURAL := 0;
        inst_stages : POSITIVE := 3;
        inst_out_reg : NATURAL := 0;
        inst_tc_mode : NATURAL := 0;
        inst_rst_mode : NATURAL := 0;
        inst_op_iso_mode : NATURAL := 0
    );
    port (
        inst_clk : in std_logic;
        inst_rst_n : in std_logic;
        inst_a : in std_logic_vector(inst_a_width-1 downto 0);
        inst_b : in std_logic_vector(inst_b_width-1 downto 0);
        product_inst : out std_logic_vector(inst_a_width+inst_b_width-1 downto 0);
        inst_launch : in std_logic;
        inst_launch_id : in std_logic_vector(inst_id_width-1 downto 0);
        pipe_full_inst : out std_logic;
        pipe_ovf_inst : out std_logic;
        inst_accept_n : in std_logic;
        arrive_inst : out std_logic;
        arrive_id_inst : out std_logic_vector(inst_id_width-1 downto 0);
        push_out_n_inst : out std_logic;
        pipe_census_inst : out std_logic_vector(1 downto 0)
    );
end DW_lp_piped_mult_inst;

```

architecture inst of DW\_lp\_piped\_mult\_inst is

begin

```

-- Instance of DW_lp_piped_mult
U1 : DW_lp_piped_mult
    generic map ( a_width => inst_a_width,
                  b_width => inst_b_width,
                  id_width => inst_id_width,
                  in_reg => inst_in_reg,
                  stages => inst_stages,

```

```
        out_reg => inst_out_reg,
        tc_mode => inst_tc_mode,
        rst_mode => inst_rst_mode,
        op_iso_mode => inst_op_iso_mode )

port map ( clk => inst_clk,
          rst_n => inst_rst_n,
          a => inst_a,
          b => inst_b,
          product => product_inst,
          launch => inst_launch,
          launch_id => inst_launch_id,
          pipe_full => pipe_full_inst,
          pipe_ovf => pipe_ovf_inst,
          accept_n => inst_accept_n,
          arrive => arrive_inst,
          arrive_id => arrive_id_inst,
          push_out_n => push_out_n_inst,
          pipe_census => pipe_census_inst );

end inst;

-- Configuration for use with a VHDL simulator
-- pragma translate_off
library DW03;
configuration DW_lp_piped_mult_inst_cfg_inst of DW_lp_piped_mult_inst is
    for inst
    end for; -- inst
end DW_lp_piped_mult_inst_cfg_inst;
-- pragma translate_on
```

## HDL Usage Through Component Instantiation - Verilog

```

module DW_lp_piped_mult_inst( inst_clk, inst_rst_n, inst_a, inst_b, product_inst,
inst_launch, inst_launch_id,
    inst_accept_n, arrive_inst, arrive_id_inst, pipe_full_inst,
    pipe_ovf_inst, push_out_n_inst, pipe_census_inst );

parameter a_width = 8;
parameter b_width = 8;
parameter id_width = 8;
parameter in_reg = 0;
parameter stages = 3;
parameter out_reg = 0;
parameter tc_mode = 0;
parameter rst_mode = 0;
parameter op_iso_mode = 0;

`define census_width 2 // ceil(log2(max(1, in_reg+(stages-1)+out_reg)+1))

input inst_clk;
input inst_rst_n;
input [a_width-1:0] inst_a;
input [b_width-1:0] inst_b;
output [a_width+b_width-1:0] product_inst;
input inst_launch;
input [id_width-1 : 0] inst_launch_id;
input inst_accept_n;
output arrive_inst;
output [id_width-1 : 0] arrive_id_inst;
output pipe_full_inst;
output pipe_ovf_inst;
output push_out_n_inst;
output [`census_width-1 : 0] pipe_census_inst;

// Instance of DW_lp_piped_mult
DW_lp_piped_mult #(a_width, b_width, id_width, in_reg, stages, out_reg, tc_mode,
rst_mode, op_iso_mode)
    U1 ( .clk(inst_clk),
        .rst_n(inst_rst_n),
        .a(inst_a), .b(inst_b),
        .product(product_inst),
        .launch(inst_launch),
        .launch_id(inst_launch_id),
        .accept_n(inst_accept_n),
        .arrive(arrive_inst),
        .arrive_id(arrive_id_inst),
        .pipe_full(pipe_full_inst),
        .pipe_ovf(pipe_ovf_inst),

```

---

```
        .push_out_n(push_out_n_inst),  
        .pipe_census(pipe_census_inst)  
    );  
  
endmodule
```

## Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

| Date           | Release       | Updates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2024 | DWBB_202409.0 | <ul style="list-style-type: none"> <li>Updated date and version; no technical changes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                          |
| December 2023  | DWBB_202312.0 | <ul style="list-style-type: none"> <li>Updated date and version; no technical changes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                          |
| July 2023      | DWBB_202212.5 | <ul style="list-style-type: none"> <li>Updated version and date</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| July 2020      | DWBB_201912.5 | <ul style="list-style-type: none"> <li>Adjusted content and title of <a href="#">“Suppressing Warning Messages During Verilog Simulation”</a> on page <a href="#">7</a> and added the DW_SUPPRESS_WARN macro</li> </ul>                                                                                                                                                                                                                                                                                   |
| October 2019   | DWBB_201903.5 | <ul style="list-style-type: none"> <li>Added the “Disabling Clock Monitor Messages” section</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                    |
| March 2019     | DWBB_201903.0 | <ul style="list-style-type: none"> <li>Clarified the op_iso_mode parameter in <a href="#">Table 1-2</a> on page <a href="#">2</a></li> <li>Clarified licensing requirements in <a href="#">Table 1-3</a> on page <a href="#">3</a></li> <li>Added <a href="#">Figure 1-2</a> on page <a href="#">5</a> to clarify datapath gating</li> <li>Added <a href="#">“Enabling minPower”</a> on page <a href="#">13</a></li> <li>Added this Revision History table and the document links on this page</li> </ul> |

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